

Extended Essay

Essay Title

The Impact of Dust on the Efficiency of Solar Photovoltaic (PV) Cells

Research Question

To what extent does the accumulation of dust particles affect the efficiency of solar panels?

Subject: Physics

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1. Introduction

In response to climatic change this has stirred and inspired global initiatives to transform towards 'Cleaner Energy Sources' as one way to reduce the climatic change effects such as wind, solar and hydroelectric power (United Nations, 2023). To commit to a culture of environmental responsibility, our campus has started to transition to an energy source that is considered safe and has a minimum greenhouse gas emission namely solar energy (Ritchie, 2020). This can be achieved through the installation of solar panels, solar heaters in hostels and putting up of solar-powered security lights with the ultimate goal of disconnecting from the Eswatini National Power Grid in 2030.

The power that solar panels produce may be affected by multiple factors, especially environmental factors such as temperature and weather conditions (Seventh Queen & Kolkowska, 2023). Despite Eswatini not being densely populated, the country faces moderately unsafe air quality, marked by annual mean concentrations of PM_{2.5} (Particulate Matter of fine particles with a diameter less than 2.5µg) is 17µg/m³ which exceeds the recommended maximum of 10µg/m³ (CRT Group, 2021). This definitively means that even though dust particles constitute less than 1% of the air composition, the percentage of dust particles in the air in school is higher (Byju's, 2020). Hence, in my essay, I am investigating how dust particles may affect the efficiency of solar panels and find solutions to minimise these effects. Therefore the essay focuses on answering the big question:

To what extent does the accumulation of dust particles affect the efficiency of solar panels?

1.1. Methodology

I will conduct both primary and secondary research to effectively address my research question. My primary research will involve setting up an experiment, allowing me to tailor the electrical parameters collected to suit the conditions of the solar panel at school in Eswatini, ensuring a contextualised investigation. Additionally, secondary research will be employed to support my arguments and counterarguments. This will allow the comparison and contrast of results obtained from credible scientific sources, thereby mitigating personal bias and minimising subjectivity in addressing my research question.

1.2. Theoretical background

1.2.1. How the sun reaches the Earth

The sun serves as the primary source of energy for solar systems, emitting electromagnetic radiation across various wavelengths and frequencies, with visible light being of particular significance. This radiation energy is approximately 3.95×10^{26} Joules per second which will start my exploration into solar energy (Coddington et al., 2016). Stefan-Boltzmann's law states that the total energy radiated per unit second by one square metre of a black body is directly proportional to the fourth power of temperature (Vedantu, 2024). This can be represented as;

$$P = \sigma AT^4 \quad [1]$$

Where:

P = Power

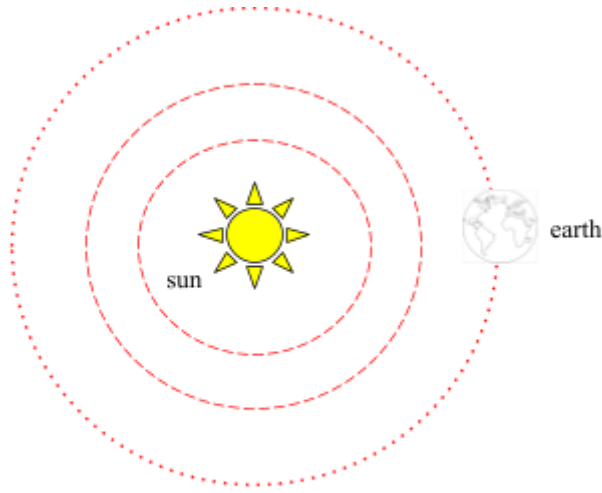
σ = Stefan-Boltzmann constant

T = Absolute temperature

$$P = 5.67 \times 10^{-8} \times 4\pi(7.0 \times 10^8)^2 \times 5800^4 \quad [2]$$

$$P = 3.95 \times 10^{26} \text{ W} \quad [3]$$

Since this sun radiation is spread over a large area to reach the earth, its power reduces significantly as intensity is inversely proportional to the area (Brownson, 2014). This can be shown as;



$$Intensity = \frac{Power}{Area} \quad [4]$$

$$Intensity = \frac{3.95 \times 10^{26} W}{4\pi(1.5 \times 10^{11} m)^2} \quad [5]$$

$$Intensity = 1397 W m^{-2} \quad [6]$$

Figure 1: Earth's orbit around the Sun

The intensity is the amount of energy received from the sun per unit time per unit area. This is known as the solar constant (Zirin, 2019). Furthermore, this solar constant is spread out over the spherical earth's surface. Dividing it by four accounts for the hemisphere facing the Sun, as the Earth is exposed to only half the Sun's radiation at any time due to its shape and rotation (Munn, 2004, pp. 13–30).

$$Intensity = \frac{1397 W m^{-2}}{4} = 39.25 W m^{-2} \quad [7]$$

The earth's terrestrial albedo, approximately 0.3, is multiplied by this intensity resulting in an adjusted intensity that reaches the planet's surface to be $0.7 \times 349 = 244$ Watts per m^2 . This intensity represents the solar energy that ideally reaches the earth's surface excluding effects such as cloud cover and which serves as the basis for solar power generation in solar panels and to answer the big question.

1.2.2. Solar Photovoltaic Cell Operations

There are two types of solar panels which include thermal and photovoltaic panels. PV modules in function by harnessing solar radiation and converting sunlight into electricity through the photoelectric effect. The PV modules are made up of semiconductor materials such as silicon. Semiconductors act as insulators at low temperatures and in slightly higher temperatures, conditions which the sun can satisfy, therefore act as conductors. Conduction will occur in higher temperatures as electrons will be knocked by light photons from the sun in the p-region that can

break away from their covalent bonds and move freely about their crystal lattice. This creates holes that ‘move’ as a result of electrons filling neighbouring holes in the n-region. With terminals connected and electron drift > hole diffusion in the p-region and the Hole drift is < electron diffusion, extra carriers flow through generating electricity in the PV module (Honsberg & Bowden, 2019).

1.2.3. The role of dust in solar photovoltaic cells

As sunlight is needed to produce electricity in the solar panel being an electromagnetic wave it will also interact with particles that will be present on its surface (e.g. dust particles) which range from individual electrons to even larger particles. When illuminated by the electromagnetic wave, the electric charges within the dust particles are set into oscillatory motion by the electric field of the incident wave. This motion results in the re-radiation of electromagnetic energy in all directions, a phenomenon known as scattering, particularly, elastic scattering (Mann et al., 2009). In addition to scattering, excited charges in dust particles transform some of the incident electromagnetic energy into thermal energy in a process known as absorption (Bohren & Huffman, 2009). Scattering and absorption combined effects are known as extinction which is the total energy loss of the incident electromagnetic wave which here is solar radiation. This can be expressed as:

$$\sigma_{extinction} = \sigma_{scattering} + \sigma_{absorption}$$

[8]

Dust particles act as obstacles that scatter and absorb incoming sunlight, reducing the amount of energy reaching the solar cells. As a result, the efficiency of the solar panel decreases because less sunlight is converted into electricity. Following these properties, dust particles on the surface of the solar panel can scatter incoming light in various directions, including back toward the panel itself. This reduces the amount of sunlight reflected away from the solar panel effectively increasing the solar energy absorbed by the panel and consequently leading to a rise in temperature within PV cells.

1.2.4. How dust accumulation raises solar cell temperature affecting panel efficiency

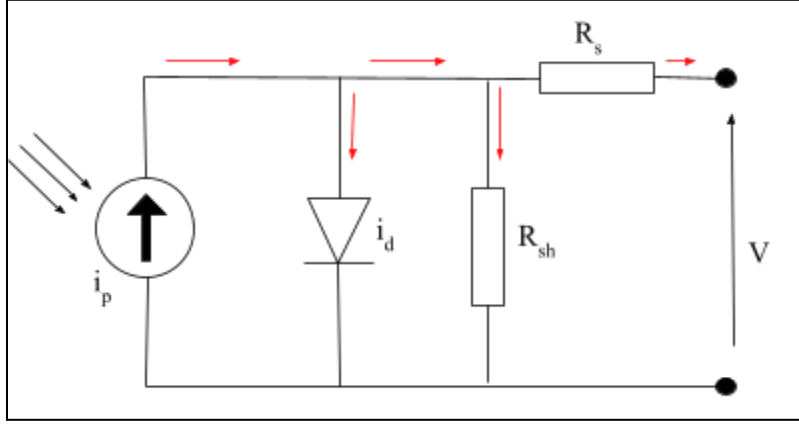


Figure 2: Solar cell equivalent circuit diagram

This is a model of a solar cell which are the cells made up of the panel. The two main characteristics which temperature that may affect the efficiency of the solar panel are: current and voltage. The photovoltaic cells produce three types of current which can be represented by the following equation:

$$I = I_p - I_d - I_{R_{sh}}$$

[9]

Where:

I = Total Current

I_p = Photocurrent;

I_d = Diode current;

$i_{R_{sh}}$ = Current going through the shunt resistor;

I_d is the current going through the diode which can be represented as $I_d = I_o \left(e^{\frac{V+iR_s}{nV_t}} - 1 \right)$ (Martí

et al., 1997). The current passing through the resistor R_{sh} can be represented as $\frac{V+iR_s}{R_{sh}}$. We can

therefore substitute the values into Equation 9 to get;

$$i = i_p - I_o \left(e^{\frac{V+iR_s}{nV_t}} - 1 \right) - \frac{V+iR_s}{R_{sh}}$$

Where;

I_o = Reverse saturation current;

V = Voltage;

R_s = Resistor connected in series;

N = Diode ideality factor;

V_T = Terminal voltage;

R_{sh} = Shunt resistance.

1.2.5. Establishing the relationship between current and temperature

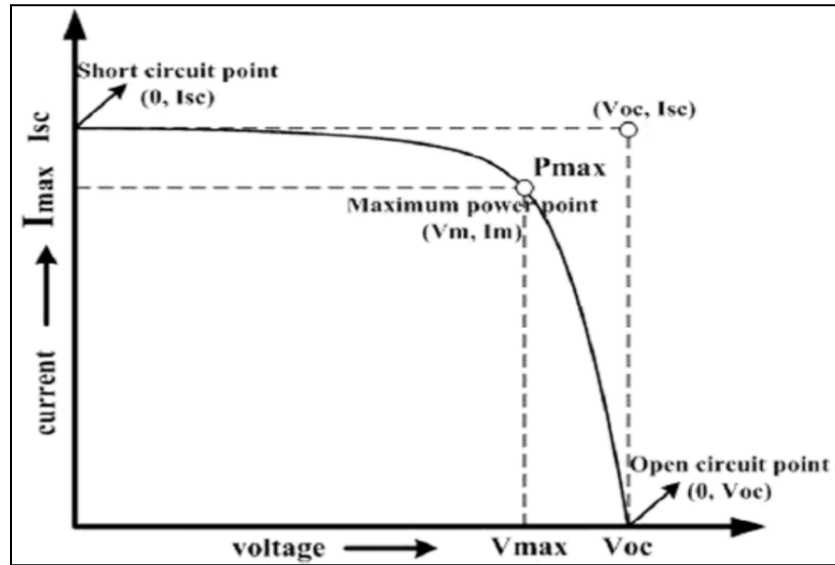


Figure 3: I - V Characteristics of a solar cell

Note: By (El-Ahmar et al., 2016). A typical V - I characteristic of a solar cell

When I is maximum, the voltage is at 0 which means the terminal voltage has been short-circuited, therefore at that point $V = 0$ and $I = \text{maximum}$, we can replace V and put 0 to get;

$$I = I_p - I_o \left(e^{\frac{0 + IR_s}{nV_T}} - 1 \right) - \frac{0 + IR_s}{R_{sh}}$$

As R_s is tending towards 0 and is very negligible we can remove it from the equation and get that short-circuit current is equal to photocurrent. The photocurrent is directly proportional to insolation (Ramgolam & Soyjaudah, 2018). Through this relationship, we can therefore see that an increase in temperature leads to a corresponding increase in photocurrent. Since $i_{sc} \approx i_p$, this means short circuit current increases. Due to the reduction in band gap energy, the energy required for an electron to transition into the conduction band decreases, allowing more valence electrons to make this transition. Therefore, there is an increase in free electrons, leading to higher photocurrent and consequently greater short-circuit current (O'Donnell & Chen, 1991).

1.2.6. Establishing the relationship between voltage and temperature

When $V = \text{maximum}$, I is usually 0 meaning the shunt resistance is greater than the open circuit voltage (ie $R_{sh} > V_{OC}$) therefore we can replace these values into Equation 10 and get;

$$0 = I - I_o \left(e^{\frac{0 + iR_s}{nV_T}} - 1 \right) \quad [12]$$

Rearranging the formula to make V_{OC} the subject of the formula we get;

$$V_{OC} = \eta V_T \cdot \ln \left(\frac{I_p + I_o}{I_o} \right) \quad [13]$$

I_o is smaller compared to I_p therefore reverse saturation current can be considered negligible making the equation;

$$V_{OC} = \eta V_T \cdot \ln \left(\frac{I_p}{I_o} \right)$$

We can replace $I_o = KT^m e^{-\frac{V_{GO}}{\eta V_T}}$ it by manipulating the diode equation to get;

$$\ln(I_o) = m \cdot \ln(KT) - \frac{V_{GO}}{\eta V_T} \quad [14]$$

Since $V_T = \frac{KT}{q}$

Where;

K = Boltzmann's Constant (1.38 x 10⁻²³ J / K) ;

T = Temperature;

q = Charge of an Electron (1.6 x 10⁻¹⁹ C);

$$\begin{aligned} \ln(I_o) &= m \cdot \ln(KT) - \frac{V_{GO}}{\eta(\frac{T}{11600})} \\ \frac{d}{dt} \ln(I_o) &= \frac{m}{KT} \cdot k - \left(\frac{V_{GO}}{\eta T(\frac{T}{11600})} \right) \\ \ln(I_o) &= \frac{m}{T} + \frac{V_{GO}}{\eta T V_T} \end{aligned}$$

[15]

From Equation 13, we get;

$$\begin{aligned} \frac{V_{oc}}{\eta V_T} &= \ln\left(\frac{i_p}{I_o}\right) \\ \frac{V_{oc}}{\eta V_T} &= \ln(I_p) - \ln(I_o) \end{aligned}$$

[16]

Taking $\frac{d}{dt}$ both sides;

$$\frac{d}{dt} \left(\frac{V_{oc}}{\eta V_T} \right) = \frac{d}{dt} [\ln(I_p)] - \frac{d}{dt} [\ln(I_o)]$$

[17]

Since $\ln(I_p)$ does not change that much therefore it tends towards 0 giving us;

$$\frac{d}{dt} \left(\frac{V_{oc}}{\eta V_T} \right) = - \frac{d}{dt} [\ln(I_o)]$$

$$\frac{1}{\eta} \left[\frac{\left(V_T \cdot \frac{d}{dt} V_{oc} \right) - \left(V_{oc} \cdot \frac{d}{dt} V_T \right)}{V_T \cdot V_T} \right] = - \frac{d}{dt} [\ln(I_o)]$$

$$\frac{\frac{d}{dt} V_{oc}}{\eta V_T} - \frac{V_{oc}}{11600 \cdot \eta \cdot \frac{T}{11600} \cdot V_T} = - \frac{d}{dt} [\ln(I_o)]$$

Using Equation 15, we substitute and get;

$$\begin{aligned}\frac{\frac{d}{dt} V_{OC}}{\eta V_T} - \frac{V_{OC}}{11600 \eta \frac{T}{11600} V_T} &= - \left[\frac{m}{T} + \frac{V_{GO}}{\eta T V_T} \right] \\ \frac{d}{dt} V_{OC} \cdot T &= V_{OC} - m \eta V_T + V_{GO} \\ \therefore \frac{d}{dt} V_{OC} &= \frac{V_{OC} - m \eta V_T + V_{GO}}{T}\end{aligned}$$

[18]

To understand the relationship better from the equation, I couldn't tell the relationship directly therefore I used a linear first-order differential equation in the form of $\frac{dy}{dx} + Py = Q$.

$$\text{Where; } y = V_{OC}; x = T; P = \left(-\frac{1}{T}\right); Q = \left(\frac{V_{GO}}{T} - \frac{m \cdot \eta}{11600}\right)$$

$$\frac{d}{dt} V_{OC} + \left(\frac{-1}{T}\right) V_{OC} = -\frac{m \cdot \eta}{11600} + \frac{V_{GO}}{T}$$

[19]

$$\text{I then got the Integrating factor where } e^{\int P dx} \Rightarrow e^{\int -\frac{1}{T} dT} \Rightarrow e^{-\ln(T)} = \frac{1}{T}$$

Therefore getting; y (*Integrating Factor*) = $\int Q(\text{Integrating Factor}) dx$

$$V_{OC} \left(\frac{1}{T}\right) = \int \left(\frac{V_{GO}}{T} - \frac{m \cdot \eta}{11600}\right) \cdot \frac{1}{T} dT$$

$$V_{OC} = T \left[\int \left(\frac{V_{GO}}{T^2}\right) dT - \int \frac{m \cdot \eta}{11600} \cdot \frac{1}{T} dT \right]$$

$$V_{OC} = T \left[\left(-\frac{V_{GO}}{T^2}\right) - \left(\frac{m \cdot \eta}{11600} \cdot \ln T\right) \right] + c$$

$$V_{OC} = - \left[V_{GO} + \frac{m \cdot \eta}{11600} \cdot T \ln T \right] + c$$

[20]

This proves that an increase in temperature leads to a corresponding decrease in open circuit voltage (electromotive force).

1.3. Hypothesis

The accumulation of dust particles on the solar panel surface will result in a corresponding decrease in the efficiency of the solar panel. This is because dust creates a quasi-insulator layer which prevents the escape of reflected light rays. Consequently, this increases the temperature of the PV cells as the dust particles convert some of the incident solar rays into thermal energy. The increased temperature in turn causes a reduction in the efficiency of the cells due to reduced open circuit voltage as seen in Equation 20.

2. Experimental Description

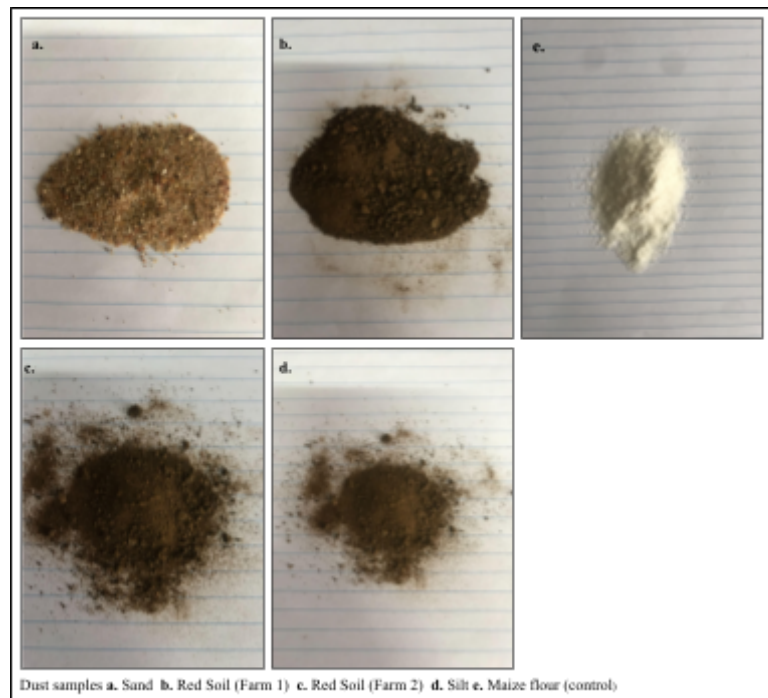
2.1. Variables

2.1.1. Independent variable

1. Mass of dust particles (g)
2. Type of dust particles

2.1.2. Dependent variable

1. Efficiency of solar panel (η)



2.1.3. Controlled variables

Variable	Control Measure	Justification
Wind	Close windows	The greater the air movement, the more dust particles will not settle on the surface of the solar panel and the less accurate my findings will be.
Amount of light in the room	The experiment was done in a dark room having a 250W incandescent bulb as the only light source	The more light in the room the more difficult it could be to measure the total amount of light hitting the solar panel.

2.2. Preparation of experimental setup

For the experiment, I used a photovoltaic monocrystalline solar panel with a maximum power rating of 20W. Electrical characteristics such as voltage in Volts (V) and current in Ampere (A), were measured and recorded. To quantify the effect of dust particles, the efficiency of the solar was compared under different dust conditions and an ideal dust-free clean condition.

Maize flour (particle E) was used as a control in my experiment because it lacks soil minerals and organic matter, providing a baseline measurement for comparison in my experiment. Furthermore, including maize flour highlights the difference of the voltage between organic (soil) and non-organic materials (maize flour).

The experiment was carried out under controlled laboratory conditions to eliminate external influences and safety precautions due to the presence of lightning. Room temperature was carefully monitored using a graduated thermometer within a dark environment, with the only source of light being a 250W incandescent light bulb. The bulb is to be positioned at a right angle with the solar panel to ensure maximum light intensity. This setup is aimed at optimising the incident light's intensity, considering its vector nature, where maximum intensity is achieved when the angle of incidence is perpendicular to the solar panel.

$$\text{Maximum Intensity} = \text{Initial Intensity} \times \cos(0^\circ)$$

$$\text{Maximum Intensity} = \text{Initial Intensity} \times 1$$

$$\therefore \text{Maximum Intensity} = \text{Initial Intensity}$$

To quantify the performance of a solar panel, efficiency can be used to calculate that. Efficiency is the ratio of energy output from the solar cell to the solar input energy.

$$\text{Efficiency } (\eta) = \frac{\text{power output}}{\text{power input}} \quad [21]$$

$$\text{Since, } \text{power output} = V_{oc} I_{sc} FF$$

Where;

V_{oc} = open-circuit voltage

I_{sc} = short-circuit current

FF = fill factor

We can rearrange Equation 21 to get;

$$\eta = \frac{V_{oc} I_{sc} FF}{\text{power input}} \quad [22]$$

2.3. Safety and environmental implications

- Ensure the use of masks to prevent inhalation of dust particles, especially while dispersing soil particles on the solar panel.
- Use safety goggles to shield the eyes from dust particles to minimise the risk of eye irritation,

- Wear gloves when handling the bulb as it gets very hot minimising the risk of burns.
- Work on a dry working surface and keep liquids away from the working space to prevent electrical shocks.
- For the soil collection, I ensured to refill any holes dug for soil collection to preserve the beauty of the landscape and prevent potential hazards.
- Any macroscopic organism found in the soil was returned to its natural habitat to respect local ecosystems and promote biodiversity. The soil was then dry and sieved to remove rocks and small debris to promote accurate experimentation and minimise contamination risks.

3. Experimental Conduction

3.1. Theoretical setup

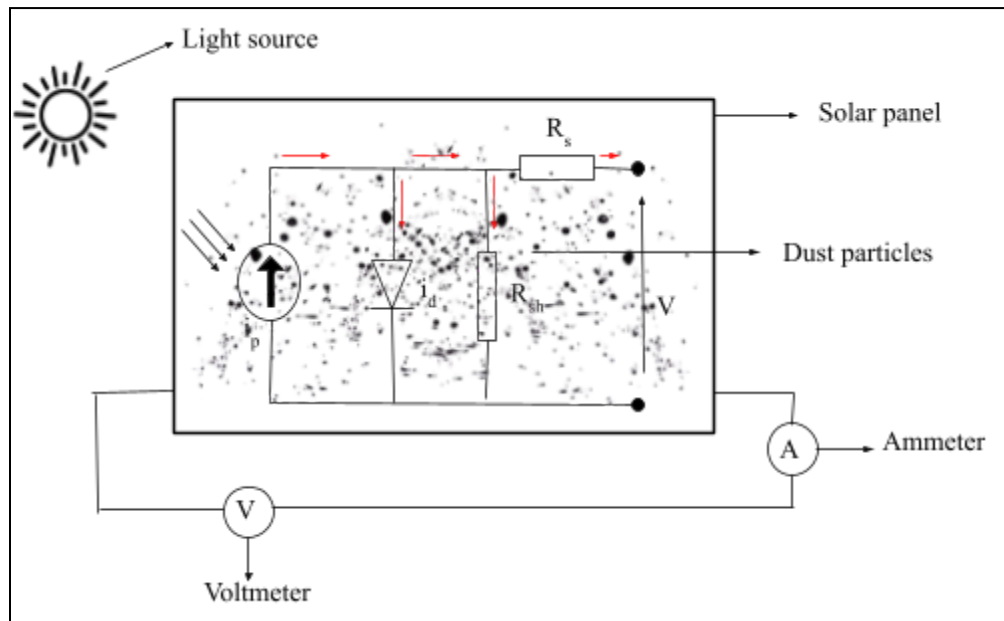


Figure 3: Theoretical Framework For Experimental Design

3.2. Apparatus

1. Metre ruler
2. Light bulb

3. Clamp and stand
4. Power source
5. Solar panel
6. Connecting wires
7. Multimeter
8. Spatula
9. Electronic balance
10. Soil samples
11. Salt shaker
12. Stopwatch
13. Protractor

3.3. Experimental setup

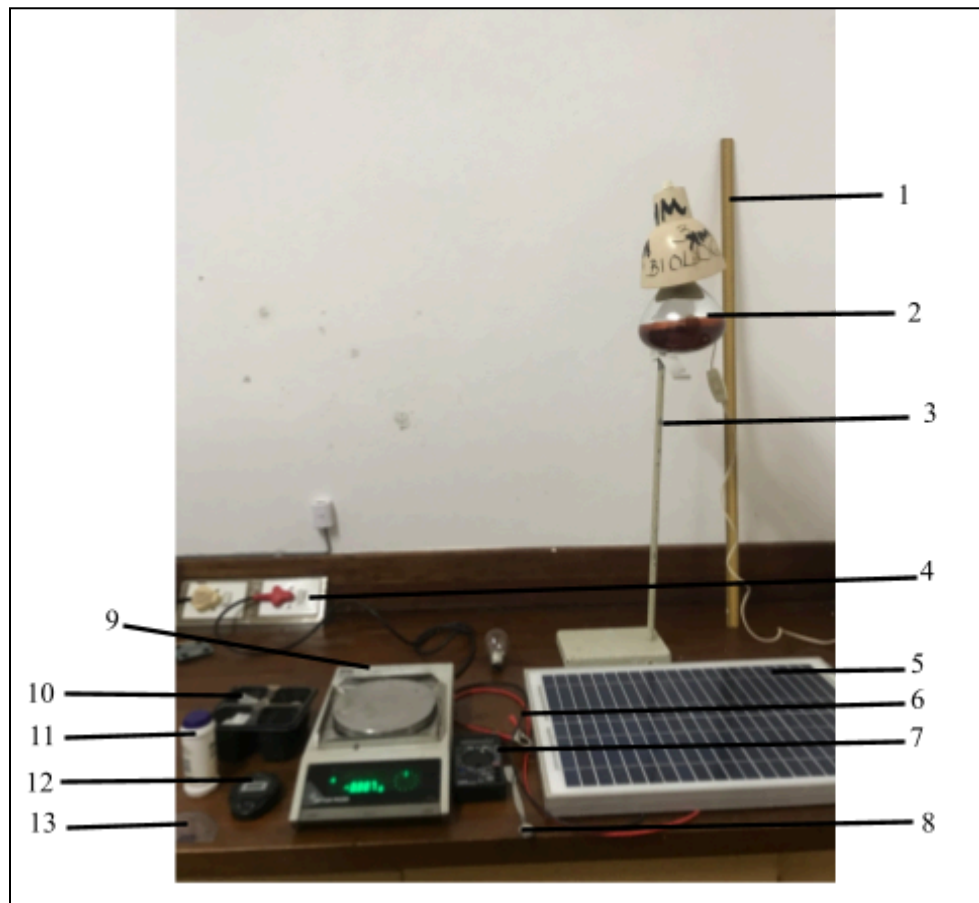


Figure 4: Practical Configuration of Experimental Apparatus

3.4. Experimental method

- Place the panel on a level surface and ensure the angle of the solar panel relative to the surface is 0 Degrees.
- Place the incandescent lightbulb at a distance of 60.4 cm above the panel and connect it to a stable power source.
- Connect the solar panel to the multimeter using well-insulated wires.
- Measure and record three values of voltage and current of the solar panel in intervals of 10 minutes using the stopwatch. Calculate the average voltage from the three recorded values to obtain the final voltage value.
- Connect the electronic balance to a reliable power source.
- Place a tissue on the electronic balance and calibrate it to 0.00 grams.
- Using a spatula, carefully collect 2.00 grams of Soil Type A and place it on the tissue on top of the electronic balance.
- Transfer the measured soil to a salt shaker using the tissue and seal it tightly to disperse the soil evenly over the solar panel using the salt shaker maintaining a distance less than 4.0cm from the panel's surface to minimise particles from drifting away from the panel.
- Measure and record the new voltage and current readings.
- Replicate the procedure by adding 2.00 grams of Soil Type A each time to measure the effects at 2.00, 4.00, 6.00, 8.00, and 10.00 grams.
- After completing the soil application, carefully collect the soil particles from the panel into a tray to prevent spillage.
- Clean the solar panel and the salt shaker using soap and water, then dry them thoroughly.
- Replicate the procedure using Soil Types B, C, D, and Maize Flour.

3.5. Experimental Results

3.5.1. Raw data

Soil Type A (Sand)					Soil Type B (Red soil from Farm 1)				
Mass of sand ± 0.01 (kg)	Voltage ± 0.01 (V)			Current ± 0.01 (A)	Mass of sand ± 0.01 (kg)	Voltage ± 0.01 (V)			Current ± 0.01 (A)
	Trial 1	Trial 2	Trial 3			Trial 1	Trial 2	Trial 3	
0.002	21.500	21.500	21.400	0.060	0.002	21.200	21.200	21.200	0.060
0.004	21.200	21.100	21.100	0.060	0.004	21.100	21.000	21.100	0.060
0.006	20.900	20.800	20.900	0.060	0.006	20.900	20.900	21.000	0.060
0.008	20.400	20.300	20.400	0.060	0.008	20.800	20.800	20.800	0.060
0.010	20.100	20.000	20.200	0.060	0.010	20.200	20.200	20.200	0.060

Soil Type C (Red soil from Farm 2)					Soil Type D (Silt)				
Mass of sand ± 0.01 (kg)	Voltage ± 0.01 (V)			Current ± 0.01 (A)	Mass of sand ± 0.01 (kg)	Voltage ± 0.01 (V)			Current ± 0.01 (A)
	Trial 1	Trial 2	Trial 3			Trial 1	Trial 2	Trial 3	
0.002	21.500	21.500	21.600	0.060	0.002	21.700	21.700	21.700	0.060
0.004	21.200	21.100	21.100	0.060	0.004	21.400	21.400	21.400	0.060
0.006	21.100	21.000	21.000	0.060	0.006	21.200	21.200	21.100	0.060
0.008	21.000	21.000	21.100	0.060	0.008	21.000	21.000	21.000	0.060
0.010	20.900	20.900	20.900	0.060	0.010	21.000	19.800	19.800	0.060

Control Particles E (Maize Flour)				
Mass of sand ± 0.001 (kg)	Voltage ± 0.01 (V)			Current ± 0.01 (A)
	Trial 1	Trial 2	Trial 3	
0.002	20.800	20.800	20.800	0.050
0.004	20.600	20.600	20.500	0.040
0.006	20.400	20.300	20.400	0.040
0.008	20.200	20.200	20.200	0.040
0.010	19.800	19.900	19.900	0.030

Figure 5: Representation of Raw Data (See Appendix A, B, C, D, E)

3.5.2. Calculated Data

Table 1: Average Voltage and Uncertainty for Soil Type A Across Different Masses

Soil Type A (Sand)						
Mass of sand (kg)	Voltage			Average Voltage	Uncertainty of voltage	Current
	Trial 1	Trial 2	Trial 3			
0.002	21.500	21.500	21.400	21.467	0.050	0.060
0.004	21.200	21.100	21.100	21.133	0.050	0.060
0.006	20.900	20.800	20.900	20.867	0.050	0.060
0.008	20.400	20.300	20.400	20.367	0.050	0.060
0.010	20.100	20.000	20.200	20.100	0.100	0.060
0.012	19.900	19.900	20.000	19.933	0.050	0.060
0.014	19.700	19.700	19.700	19.700	0.000	0.060
0.016	19.600	19.500	19.600	19.567	0.050	0.060
0.018	19.300	19.400	19.400	19.367	0.050	0.060
0.020	19.100	19.100	19.200	19.133	0.050	0.060

Table 2: Average Voltage and Uncertainty for Soil Type B Across Different Masses

Soil Type B (Red soil from Farm 1)						
Mass of sand (kg)	Voltage			Average voltage	Uncertainty of voltage	Current
	Trial 1	Trial 2	Trial 3			
0.002	21.200	21.200	21.200	21.200	0.000	0.060
0.004	21.100	21.000	21.100	21.067	0.050	0.060
0.006	20.900	20.900	21.000	20.933	0.050	0.060
0.008	20.800	20.800	20.800	20.800	0.000	0.060
0.010	20.200	20.200	20.200	20.200	0.000	0.060
0.012	20.000	20.000	20.100	20.033	0.050	0.060
0.014	19.900	19.800	19.800	19.833	0.050	0.060
0.016	19.600	19.600	19.600	19.600	0.000	0.060
0.018	19.500	19.500	19.400	19.467	0.050	0.060
0.020	19.400	19.300	19.300	19.333	0.050	0.060

Table 3: Average Voltage and Uncertainty for Soil Type C Across Different Masses

Soil Type C (Red soil from Farm 2)						
Mass of sand (kg)	Voltage			Average voltage	Uncertainty of voltage	Current
	Trial 1	Trial 2	Trial 3			
0.002	21.500	21.500	21.600	21.533	0.050	0.060
0.004	21.200	21.100	21.100	21.133	0.050	0.060
0.006	21.100	21.000	21.000	21.033	0.050	0.060
0.008	21.000	21.000	21.100	21.033	0.050	0.060
0.010	20.900	20.900	20.900	20.900	0.000	0.060
0.012	20.700	20.800	20.800	20.767	0.050	0.060
0.014	20.500	20.500	20.500	20.500	0.000	0.060
0.016	20.400	20.300	20.300	20.333	0.050	0.060
0.018	20.200	20.300	20.200	20.233	0.050	0.060
0.020	20.000	20.100	20.000	20.033	0.050	0.060

Table 4: Average Voltage and Uncertainty for Soil Type D Across Different Masses

Soil Type D (Silt)						
Mass of sand (kg)	Voltage			Average Voltage	Uncertainty of voltage	Current
	Trial 1	Trial 2	Trial 3			
0.002	21.700	21.700	21.700	21.700	0.000	0.060
0.004	21.400	21.400	21.400	21.400	0.000	0.060
0.006	21.200	21.200	21.100	21.167	0.050	0.060
0.008	21.000	21.000	21.000	21.000	0.000	0.060
0.010	20.000	19.800	19.800	19.867	0.100	0.060
0.012	19.600	19.700	19.700	19.667	0.050	0.060
0.014	19.500	19.500	19.500	19.500	0.000	0.060
0.016	19.400	19.400	19.300	19.367	0.050	0.060
0.018	19.200	19.200	19.200	19.200	0.000	0.060
0.020	19.000	19.100	19.000	19.033	0.050	0.060

Table 5: Average Voltage and Uncertainty for Soil Type E Across Different Masses

Particles E (Maize Flour)						
Mass of flour (kg)	Voltage			Average voltage	Uncertainty of voltage	Current
	Trial 1	Trial 2	Trial 3			
0.002	20.800	20.800	20.800	20.800	0.000	0.050
0.004	20.600	20.600	20.500	20.567	0.050	0.040
0.006	20.400	20.300	20.400	20.367	0.050	0.040
0.008	20.200	20.200	20.200	20.200	0.000	0.040
0.010	19.800	19.900	19.900	19.867	0.050	0.030
0.012	19.700	19.700	19.700	19.700	0.000	0.030
0.014	19.600	19.600	19.500	19.567	0.050	0.030
0.016	19.300	19.400	19.300	19.333	0.050	0.030
0.018	19.100	19.100	19.100	19.100	0.000	0.030
0.020	18.900	18.900	18.900	18.900	0.000	0.030

Example Calculations

1. Calculation of Average Voltage For Soil Type E

$$\text{Average Voltage} = \frac{\text{Trial 1} + \text{Trial 2} + \text{Trial 3}}{3}$$

$$\text{Average Voltage} = \frac{20.800 + 20.800 + 20.800}{3} = 20.800V$$

2. Calculation of Uncertainty of Voltage For Soil Type E

$$\Delta \text{ of Voltage} = \frac{(\text{Maximum Voltage} - \text{Minimum Voltage})}{2}$$

$$\Delta \text{ of Voltage} = \frac{(20.800 - 20.800)}{2} = 0.000V$$

3.5.3. Graphical representation

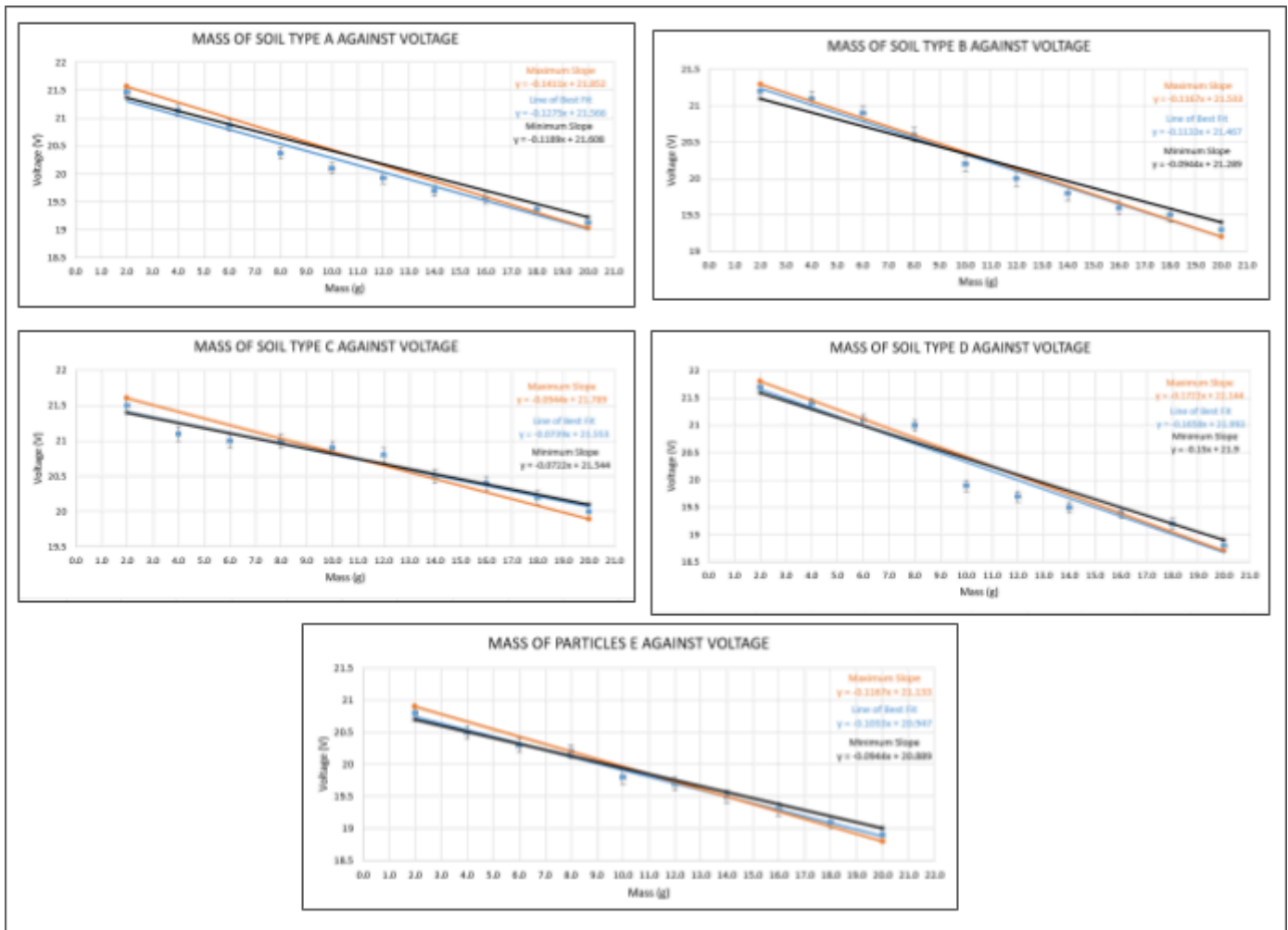


Figure 6: Mass of Soils (A, B,C ,D, E(control)) Against Voltage

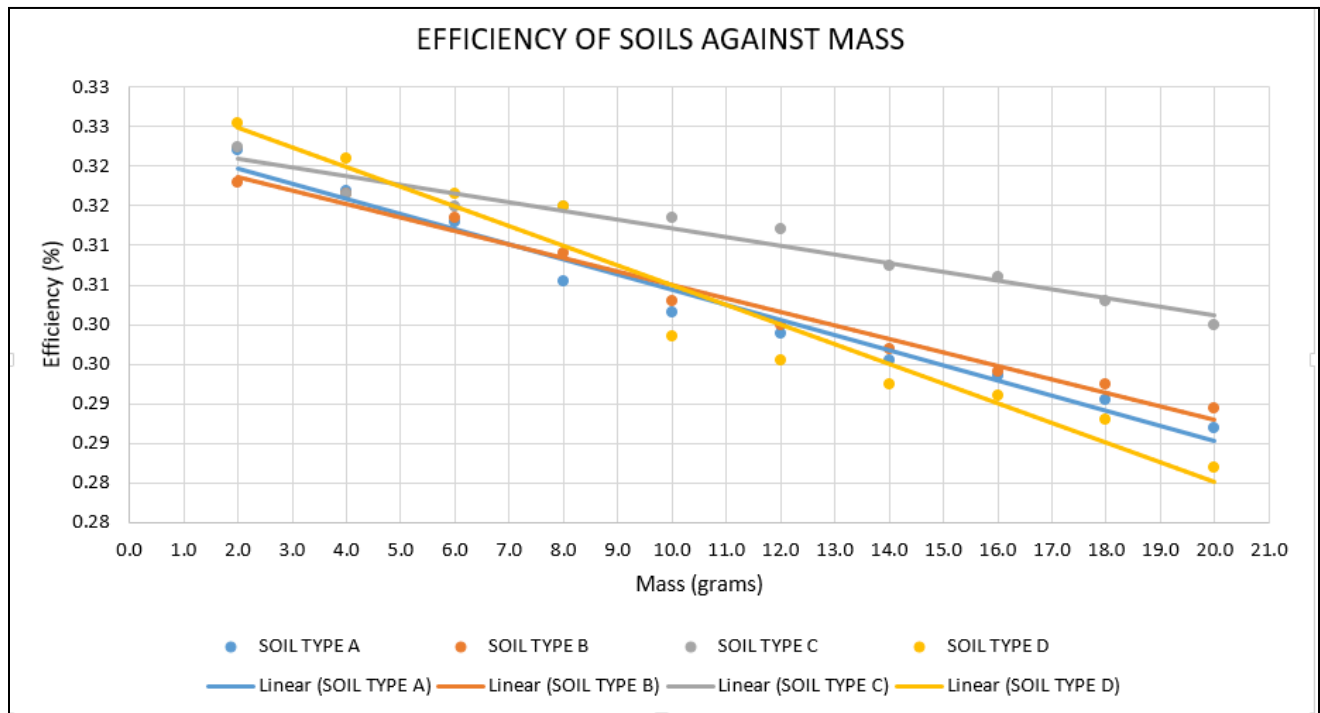


Figure 7: Efficiency of the solar panel against Soil Mass

4. Argumentation

The data I collected agrees with my initial experimental hypothesis which states that an increase in soil particles will lead to a corresponding decrease in the efficiency of the solar panel. The graphs represent mass in grams against Voltage in Volts, since V is directly proportional to efficiency, a decrease in voltage will lead to a corresponding decrease in the efficiency of the solar panel which is proven by both Figure 6 and Figure 7.

Efficiency was calculated at different soil mass increments, revealing a systematic linear decline across varying soil compositions. The graphs produced linear functions where the y-axis represents efficiency, x-axis represents the soil masses and m which is the gradient of the equation represents efficiency per unit mass of the soil.

Silt demonstrated the least effect on the efficiency of the solar panel, followed by sand, red soil sourced from farm 1, maize flour particles and red soil obtained from farm 2, in descending order. The lower efficiency observed with silt must be attributed to its finer particle size and smoother texture which results in fewer shading effects on the solar panel surface. Additionally,

sand particles have a noticeable light reflectivity compared to silt, allowing for less absorption of sunlight by the solar panel's photovoltaic cells hence explaining why it has a steep gradient (Pan et al., 2023).

Soil particles D has the the steepest gradient of mass against voltage with its equation of $y = -0.1658x + 21.993$ which represents a linear relationship between the mass of red soil from farm 2 and the voltage output. With a negative gradient of -0.1658, the equation illustrates that for every increase in the mass of soil particles in grams on the solar panel, the voltage output decreases by 0.1658 volts.

Using the model modifications for my solar panel;

Model	Rocksonic RSP-6020-X1
Maximum power	20 W
Open circuit voltages	25.4 V
Short circuit currents	1.26 A
Maximum power voltages	17.2 V
Maximum power currents	1.16 A
Module dimensions	530 x 355 x 28 mm

$$\eta = \frac{25.4 \times 1.26 \times \frac{5000}{8001}}{250} \times 100\% \quad [21]$$

$$\eta = 0.08 \times 100\% \quad [22]$$

$$\eta = 8\%$$

This means that only 8% of the sunlight's input is successfully converted into usable electric energy, meaning 92% percent is either reflected, absorbed by the material but not converted into electricity or lost in the form of heat energy. In addition to this, dust particles influence the efficiency of the solar panel. In the case of polycrystalline solar panels, an obstruction such as

dust particles or bird droppings prevents the panel from working effectively as the solar cells are connected and wired in a series connection. This is a big disadvantage as if one cell is covered by dust particles, the entire solar panel is affected which reduces the effective surface area of the solar panel receiving light.

Surprisingly, as illustrated in Figure 2, the application of different dust particles on the solar panel led to a 96.13% decline in efficiency, signifying a very great negative impact on its performance. This big reduction highlights the influence of environmental factors on the efficiency of a solar panel, especially in hilly areas. Consequently, the solar panel highlights efficiency levels ranging from 0.33% to 0.29% under different soil types and masses showing the notable variability in performance under different environmental conditions.

This implies that solar panels installed in agricultural areas, where soil particles and dust are more due to farming activities, such as where my school is situated, are likely to accumulate dirt faster per unit time than ones in the city. Agricultural activities such as bush fallowing and harvesting generate an increase in dust and particulate matter in the atmosphere which may settle on solar panels, impacting their efficiency.

5. Counter Arguments and Responses

5.1. Counter Arguments (objections to my position)

Objection 1: Dust accumulation may have negligible effects of solar panel efficiency

Researchers argue that the impact of dust accumulation on solar panel efficiency is minimal and does not warrant significant power losses. They suggest that periodic cleaning can minimise any potential decrease in efficiency due to dust accumulation as some may be carried away with air action such as wind (Adnan Shariah & Ehab Al-Ibrahim, 2023).

5.2. Reasons to Refute Counter Arguments

Refutation of Objection1:

While it is true that dust accumulation may initially have a minor impact on solar panel efficiency, the cumulative effect over time is substantial. This is very evident in my data

collection where the efficiency of solar panels is reduced from 8% to an average of 0.31% which is very high. Research studies have also shown that even a thin layer of dust can reduce solar panel efficiency up to a 50% decrease which is also very high making that counterargument invalid as per my research and the research papers I went through (Sudhakar & Dayal Singh Rajput, 2013). Additionally, the effects of dust on solar panels can be minimised or increased depending on the geographical location. Research shows that solar panels in hot and dryer regions are more likely to be affected than in cool and wet regions (Nezamisavojbolaghi et al., 2023).

6. Evaluation

Strengths

- Conduction of the experiment in a controlled environment ensures consistency by eliminating confounding variables such as wind that could influence soil composition.
- Optimisation of light intensity. Positioning the light source at a right angle maximizes light intensity, showing the maximum effect of dust on solar panel efficiency.

Weaknesses

- The use of an artificial light source. This may not accurately mimic real-world conditions as solar panels are exposed to varying levels of sunlight intensity throughout the day.
- Fluctuations in room temperature. The experiment was conducted in a dark room, however, maintaining room temperature was not possible due to lack of heating or cooling systems in the room. This could have led to fluctuations in my data affecting the validity of my findings.

Improvements

- Using natural sunlight to provide a more realistic representation of the real-world conditions, enhancing the ecological validity of my experimental findings.

7. Conclusion

For this investigation to find how dust accumulation affects the efficiency of solar PV cells, an experimental model of a solar panel model using different types and masses of dust particles was used in order to calculate efficiency before and after. Hence the answer to my research question states:

To what extent does the accumulation of dust particles affect the efficiency of solar panels?
is found to be as high as a 96.13% reduction in efficiency in an 8-year-old polycrystalline solar model.

In this investigation, I was able to observe very shocking results. The accumulation of dust particles **significantly** impacts the efficiency of solar panels, leading to a huge decline in efficiency which has been recorded to be less than 1%. This demonstrates a very negative impact on solar panel performance, showing the influential role of environmental factors particularly in hilly areas. The efficiency levels of the solar panel ranged from 0.33% to 0.29% under various soil types and masses, indicating notable variability in performance under different environmental conditions.

There have been discoveries of mitigation strategies to reduce the effect of dust accumulation on solar panels. Some of the ways include regular cleaning of the surface of solar panels using soft brushes, sponges or low-pressure water jets. This method is very cost-effective as it does not need the hiring of any expert but nonetheless, the owner needs to read the manufacturer's manual to not damage the solar panel (Gupta, 2023).

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9. Appendix

Appendix A: Voltage and Current Data for Soil Type A Across Different Masses

Soil Type A (Sand)				
Mass of sand $\pm$ 0.001 (kg)	Voltage $\pm$ 0.01 (V)			Current $\pm$ 0.01 (A)
	Trial 1	Trial 2	Trial 3	
0.002	21.500	21.500	21.400	0.060
0.004	21.200	21.100	21.100	0.060
0.006	20.900	20.800	20.900	0.060
0.008	20.400	20.300	20.400	0.060
0.010	20.100	20.000	20.200	0.060
0.012	19.900	19.900	20.000	0.060
0.014	19.700	19.700	19.700	0.060
0.016	19.600	19.500	19.600	0.060
0.018	19.300	19.400	19.400	0.060
0.020	19.100	19.100	19.200	0.060

Appendix B: Voltage and Current Data for Soil Type B Across Different Masses

Soil Type B (Red soil from Farm 1)				
Mass of sand $\pm$ 0.001 (kg)	Voltage $\pm$ 0.01 (V)			Current $\pm$ 0.01 (A)
	Trial 1	Trial 2	Trial 3	
0.002	21.200	21.200	21.200	0.060
0.004	21.100	21.000	21.100	0.060
0.006	20.900	20.900	21.000	0.060
0.008	20.800	20.800	20.800	0.060
0.010	20.200	20.200	20.200	0.060
0.012	20.000	20.000	20.100	0.060
0.014	19.900	19.800	19.800	0.060
0.016	19.600	19.600	19.600	0.060
0.018	19.500	19.500	19.400	0.060
0.020	19.400	19.300	19.300	0.060

Appendix C: Voltage and Current Data for Soil Type C Across Different Masses

Soil Type C (Red soil from Farm 2)				
Mass of sand $\pm$ 0.001 (kg)	Voltage $\pm$ 0.01 (V)			Current $\pm$ 0.01 (A)
	Trial 1	Trial 2	Trial 3	
0.002	21.500	21.500	21.600	0.060
0.004	21.200	21.100	21.100	0.060
0.006	21.100	21.000	21.000	0.060
0.008	21.000	21.000	21.100	0.060
0.010	20.900	20.900	20.900	0.060
0.012	20.700	20.800	20.800	0.060
0.014	20.500	20.500	20.500	0.060
0.016	20.400	20.300	20.300	0.060
0.018	20.200	20.300	20.200	0.060
0.020	20.000	20.100	20.000	0.060

Appendix D: Voltage and Current Data for Soil Type D Across Different Masses

Soil Type D (Silt)				
Mass of sand $\pm$ 0.001 (kg)	Voltage $\pm$ 0.01 (V)			Current $\pm$ 0.01 (A)
	Trial 1	Trial 2	Trial 3	
0.002	21.700	21.700	21.700	0.060
0.004	21.400	21.400	21.400	0.060
0.006	21.200	21.200	21.100	0.060
0.008	21.000	21.000	21.000	0.060
0.010	21.000	19.800	19.800	0.060
0.012	19.600	19.700	19.700	0.060
0.014	19.500	19.500	19.500	0.060
0.016	19.400	19.400	19.300	0.060
0.018	19.200	19.200	19.200	0.060
0.020	19.000	19.100	19.000	0.060

Appendix E: Voltage and Current Data for Soil Type E Across Different Masses

Control Particles E (Maize Flour)				
Mass of sand $\pm$ 0.001 (kg)	Voltage $\pm$ 0.01 (V)			Current $\pm$ 0.01 (A)
	Trial 1	Trial 2	Trial 3	
0.002	20.800	20.800	20.800	0.050
0.004	20.600	20.600	20.500	0.040
0.006	20.400	20.300	20.400	0.040
0.008	20.200	20.200	20.200	0.040
0.010	19.800	19.900	19.900	0.030
0.012	19.700	19.700	19.700	0.030
0.014	19.600	19.600	19.500	0.030
0.016	19.300	19.400	19.300	0.030
0.018	19.100	19.100	19.100	0.030
0.020	18.900	18.900	18.900	0.030